Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Three different gases X, Y and Z of molecular masses 2,16 and 64 were enclosed in a vessel at constant temperature till equilibrium is reached. Which of the following statement is correct?

(A) Gas Z will be at the top of the vessel

(B) Gas Y will be at the top of the vessel

(C) Gas Z will be at the bottom and X will be at the top

(D) Gases will form homogenous mixture

Q.2

Equal masses of methane and oxygen are mixed in an empty container at $25^{\circ}\mathrm{C}$. The fraction of the total pressure exerted by oxygen is

(A) $1/2$

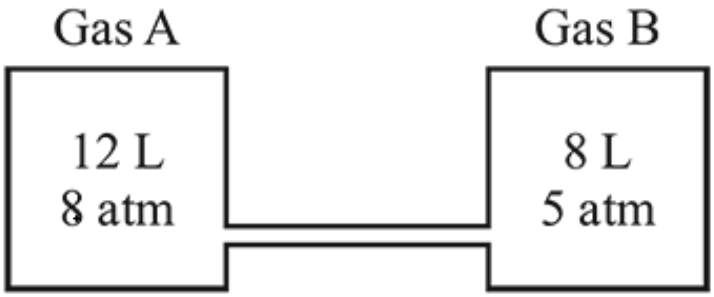
(B) $2/3$

(C) $\frac{1}{3} \times \frac{273}{298}$

(D) $1/3$

Q.3

Two vessels containing gases A and B are interconected as shown in the figure. The stopper is opened, the gases are allowed to mix homogeneously. The partial pressures of A and B in the mixture will be, respectively



(A) 8 and $5\ \mathrm{atm}$

(B) 9.6 and $4\ \mathrm{atm}$

(C) 4.8 and $2\ \mathrm{atm}$

(D) 6.4 and $4\ \mathrm{atm}$

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Q.4

Two vessels of volumes $16.4\ \mathrm{L}$ and $5\ \mathrm{L}$ contain two ideal gases of molecular existence at the respective temperature of $27^{\circ}\mathrm{C}$ and $227^{\circ}\mathrm{C}$ and exert 1.5 and 4.1 atmospheres respectively. The ratio of the number of molecules of the former to that of the later is

(A) 2

(B) 1

(C) $\frac{1}{2}$

(D) $\frac{1}{3}$

Q.5

A high- altitude balloon is filled with $1.41 \times 10^4\ \mathrm{L}$ of hydrogen at a temperature of $21^{\circ}\mathrm{C}$ and a pressure of 745 torr. What is the volume of the balloon at a height of $20\ \mathrm{km}$, where the temperature is $-48^{\circ}\mathrm{C}$ and the pressure is 63.1 torr?

(A) $1.247 \times 10^5\ \mathrm{L}$

(B) $1.66 \times 10^5\ \mathrm{L}$

(C) $1.66 \times 10^4\ \mathrm{L}$

(D) None of these

Q.6

Dimethyl ether decomposes as
$\mathrm{CH_3OCH_3(\ g)} \longrightarrow \mathrm{CH_4(\ g)} + \mathrm{CO(g)} + \mathrm{H_2(\ g)}$
when $\mathrm{CH_3OCH_3}$ decomposes to 20% extent under certain conditions, what is the ratio of diffusion of pure $\mathrm{CH_3OCH_3}$ with methane?

(A) $0.59 : 1$

(B) $1.18 : 1$

(C) $2.36 : 1$

(D) $1.77 : 1$

Q.7

A $3 : 2$ molar mixture of $\mathrm{N_2}$ and CO is present in a vessel at 500 bar pressure. Due to hole in the vessel, the gas mixture leaks out. The composition of mixture effusing out initially is

(A) $\mathrm{n_{N_2}} : \mathrm{n_{CO}} :: 1 : 2$

(B) $\mathrm{n_{N_2}} : \mathrm{n_{CO}} :: 6 : 1$

(C) $\mathrm{n_{CO}} : \mathrm{n_{N_2}} :: 1 : 2$

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(D) $n_{CO} : n_{N_2} :: 2 : 3$

Q.8

Same mass of CH_4 and H_2 is taken in container. The partial pressure caused by H_2 is

(A) $8/9$

(B) $1/9$

(C) $1/2$

(D) 1

Q.9

Equal volumes of two gases which do not react together are enclosed in separate vessels. Their pressures are $100\ mm$ and $400\ mm$ respectively. If the two vessels are joined together, then what will be the pressure of the resulting mixture (temperature remaining constant)

(A) $125\ mm$

(B) $500\ mm$

(C) $1000\ mm$

(D) $250\ mm$

Q.10

Equal masses of methane and oxygen are mixed in an empty container at $25^{\circ}C$. The fraction of the total pressure exerted by oxygen is

(A) $1/2$

(B) $2/3$

(C) $\frac{1}{3} \times \frac{273}{298}$

(D) $1/3$.

Q.11

A mixture of two gases A and B in the mole ratio $2 : 3$ is kep in a 2 litre vessel. A second $3\ L$ vessel has the same two gases in the mole ratio $3 : 5$. Both gas mixtures have the same temperature and same total pressure. They are allowed to intermix and the final temperature and the total pressure are the same as the initial values, the final volume being 5 litres. Given that the molar masses are M_A and M_B. What is the mean molar mass of the final mixture?

(A) $\frac{5M_A+8M_B}{13}$

(B) $\frac{77M_A+123M_B}{200}$

(C) $\frac{123M_A+77M_B}{250}$

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(D) $\frac{123M_A+77M_B}{150}$

Q.12

Equal moles of hydrogen and oxygen gases are placed in a container with a pin-hole through which both can escape. What fraction of the oxygen escapes in the time required for one-half of the hydrogen to escape?

(A) $1/8$

(B) $1/4$

(C) $3/8$

(D) $1/2$

Q.13

Sulphur dioxide and oxygen were allowed to diffuse through a porous partition. $20\mathrm{dm}^3$ of SO_2 diffuses through the porous partition in 60 seconds. The volume of O_2 in dm^3 which diffuses under the similar condition in 30 seconds
will be (atomic mass of sulphur $=32\mathrm{u}$):

(A) 7.09

(B) 14.1

(C) 10.0

(D) 28.2

Q.14

$56\ \mathrm{g}$ of nitrogen and $96\ \mathrm{g}$ of oxygen are mixed isothermally and at a total pressure of $10\ \mathrm{atm}$. The partial pressures of oxygen and nitrogen (in atm) are respectively

(A) 4,6

(B) 5,5

(C) 2,8

(D) 6,4

Q.15

The composition (by moles) of effused mixture after 4 effusion steps is X : 1 :Y respectively. If a gaseous mixture containing equal moles of H_2, O_2 and He is subjected to series of effusion steps. What is the value of X/Y?

(A) 4

(B) 3

(C) 2

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(D) 1

Answers & Solutions

Q.1 Answer:

Gases will form homogenous mixture

Solution:

All the gases occupy the available volume and will form homogeneous mixture.

Q.2 Answer:

$1/3$

Solution:

Let the mass of methane and oxygen $= \mathrm{m}$ **gm. Mole fraction of** $\mathrm{O_2}$

$= \frac{\text{Moles of } \mathrm{O_2}}{\text{Moles of } \mathrm{O_2} + \text{Moles of } \mathrm{CH_4}}$

$= \frac{\mathrm{m}/32}{\mathrm{m}/32 + \mathrm{m}/16} = \frac{\mathrm{m}/32}{3\,\mathrm{m}/32} = \frac{1}{3}$

Partial pressure of $\mathrm{O_2}$ $=$ **Total pressure** $\times$ **mole fraction of** $\mathrm{O_2}$**,**

$\mathrm{P_{O_2}} = \mathrm{P} \times \frac{1}{3} = \frac{1}{3}\mathrm{P}$

Q.3 Answer:

4.8 and 2 atm

Solution:

Moles of A, $(\mathrm{n_A}) = \frac{\mathrm{P_A V_A}}{\mathrm{RT}} = \frac{8 \times 12}{\mathrm{RT}} = \frac{96}{\mathrm{RT}}$

Moles of B**,** $(\mathrm{n_B}) = \frac{\mathrm{P_B V_B}}{\mathrm{RT}} = \frac{8 \times 5}{\mathrm{RT}} = \frac{40}{\mathrm{RT}}$

Total pressure $\times$ **total volume** $= (\mathrm{n_A} + \mathrm{n_B})\rangle$

RT

$$p \times (12 + 8) = \frac{1}{RT}(96 + 40)\mathrm{RT}$$
$$\mathrm{P} = 6.8$$

Partial pressure of A $=$ $\mathrm{P}\times$ **mole fraction of** A

$$= 6.8\left(\frac{96}{\mathrm{RT}} / \frac{96+40}{\mathrm{RT}}\right)$$
$$= 4.8 \text{ atm}$$

Partial pressure of $\mathrm{B} = 6.8 - 4.8 = 2$ **atm.**

Q.4 Answer:

2

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Solution:

Given conditions $\mathrm{V_1 = 16.4\ L,\ V_2 = 5\ L}$
$P_1 = 1.5\ \mathrm{atm}, \mathrm{P_2 = 4.1\ atm}$
$\mathrm{T_1 = 273 + 27 = 300\ K}$
$\mathrm{T_2 = 273 + 227 = 500\ K}$
Applying gas equation, $\frac{\mathrm{P_1\ V_1}}{\mathrm{P_2\ V_2}} = \frac{\mathrm{n_1\ T_1}}{\mathrm{n_2\ T_2}}$
$\frac{\mathrm{n_1}}{\mathrm{n_2}} = \frac{\mathrm{P_1\ V_1\ T_1}}{\mathrm{P_2\ V_2\ T_2}}$
$\therefore \frac{1.5 \times 16.4 \times 500}{4.1 \times 5 \times 300} = \frac{2}{1}$

Q.5 Answer:

$1.247 \times 10^5\ \mathrm{L}$

Solution:

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$
$V_2 = \frac{745 \times 1.41 \times 10^4 \times 225}{294 \times 63.1} = 1.274 \times 10^5\ \mathrm{L}$

Q.6 Answer:

$2.36 : 1$

Solution:

$$\frac{\mathrm{r_{CH_3OCH_3}}}{\mathrm{r_{CH_4}}} = \sqrt{\frac{\mathrm{M_{CH_4}}}{\mathrm{M_{CH_3OCH_3}}}} \times \frac{\mathrm{p_{CH_3OCH_3}}}{\mathrm{p_{CH_4}}}$$

$$= \sqrt{\frac{16}{46}} \times \frac{0.8}{0.2} = 2.36 : 1$$

Q.7 Answer:

$\mathrm{n_{CO} : n_{N_2} :: 2 : 3}$

Solution:

Molar ratio of $\mathrm{N_2}$ **and** CO **is** $3 : 2$**, i.e.,** $300\mathrm{bar}$ **and 200 bar, respectively.**
$\frac{\mathrm{n_{N_2}}}{\mathrm{n_{CO}}} = \sqrt{\frac{\mathrm{m_{CO}}}{\mathrm{mN_2}}} \times \frac{\mathrm{P_{N_2}}}{\mathrm{P_{CO}}}$
$= \frac{300}{200} = \frac{3}{2}$

Q.8 Answer:

$8/9$

Solution:

$\mathbf{N_{CH_4}}$ **= number of moles of** $\mathrm{CH_4} = \frac{\mathrm{m}}{16}$
$\mathbf{N_{H_2}}$ **= number of moles of** $\mathbf{H_2} = \frac{\mathrm{m}}{2}$

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Fraction partial pressure of H_2 is

$H_2 = \frac{n_{H_2}}{n_{H_2}+n_{CH_4}} = \frac{\frac{m}{2}}{\frac{m}{2}+\frac{m}{16}} = \frac{\frac{m}{2}}{\frac{9m}{16}} = \frac{8}{9}$

Q.9 Answer:

250 mm

Solution:

When two vessels are joined together, the volume will be doubled hence effective pressure will be halved

$P = \frac{P_1+P_2}{2} = \frac{100+400}{2} = 250 \text{ mm}$

Q.10 Answer:

$1/3$.

Solution:

Let the mass of methane and oxygen $= mgm$. **Mole fraction of** $O_2 = \frac{\text{Moles of } O_2}{\text{Moles of } O_2+\text{Moles of } CH_4}$

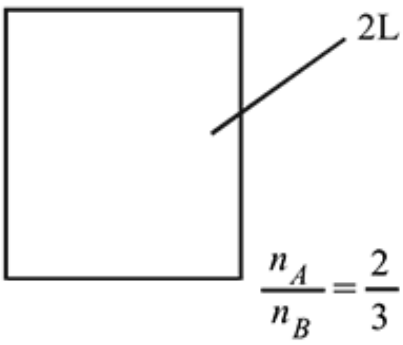
$= \frac{m/32}{m/32+m/16} = \frac{m/32}{3\,m/32} = \frac{1}{3}$

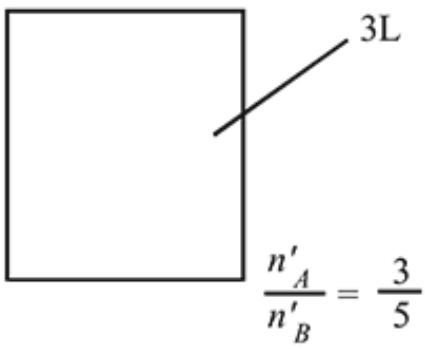
Partial pressure of O_2 **= Total pressure** $\times$ **mole fraction of** $O_2, P_{O_2} = P \times \frac{1}{3} = \frac{1}{3}P$

Q.11 Answer:

$\frac{77M_A+123M_B}{200}$

Solution:





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Mean molar mass $= \frac{\left(n_A+n'_A\right)M_A+\left(n_B+n'_B\right)M_B}{n_A+n'_A+n_B+n'_B}$

$= \frac{77M_A+123M_B}{200}$

Q.12 Answer:

$1/8$

Solution:

Given, $\mathrm{n_{H_2}} = \mathrm{n_{O_2}}$ **and** $\mathrm{t_{H_2}} = \mathrm{t_{O_2}}$

According to Graham's law of diffusion for two different gases. $\frac{\mathrm{r_{H_2}}}{\mathrm{r_{O_2}}} = \frac{\mathrm{v_1/t_1}}{\mathrm{v_2/t_2}} \Rightarrow \sqrt{\frac{\mathrm{M_{O_2}}}{\mathrm{M_{H_2}}}} = \sqrt{\frac{32}{2}}$

$\frac{1/2}{1/x} = \sqrt{16} = 4$

$\frac{x}{2} = 4$

$\therefore \mathrm{x} = 8$

$\therefore$ **Fraction of** $\mathrm{O_2} = 1/8$

Q.13 Answer:

14.1

Solution:

According to Graham's Law Diffusion:

$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}}$

Since rate of diffusion $= \frac{\text{Vol. of gas diffused } (V)}{\text{Time taken for diffusion } (t)} \therefore \quad \frac{r_1}{r_2} = \frac{V_1/t_1}{V_2/t_2}$

or $\frac{r_1}{r_2} = \frac{V_1/t_1}{V_2/t_2} = \sqrt{\frac{d_2}{d_1}}$

$= \frac{20/60}{V_2/30} = \sqrt{\frac{16/2}{32/2}} = \sqrt{\frac{1}{2}}$

$\because$ **Mol. wt** $= 2\times$ **V.D**

$\therefore \mathrm{V \cdot D} = \frac{\mathrm{Mol.wt}}{2}$

On calculating, $V_2 = 14.1$

Q.14 Answer:

6,4

Solution:

On applying Dalton's law, Partial pressure of a component = Mole fraction × Total pressure Given, mass of $\mathrm{N_2} = 56$ **g, mass of** $\mathrm{O_2} = 96$ **g**

Total pressure $= 10$ **atm**

$^{\mathrm{n}}\mathrm{N_2} = \frac{56}{28} = 2, {}^{\mathrm{n}}\mathrm{O_2} = \frac{96}{32} = 3$

$\mathrm{N_2} = \frac{{}^{\mathrm{n}}\mathrm{N_2}}{\mathrm{n}}\mathrm{N_2} + {}^{\mathrm{n}}\mathrm{O_2} = \frac{2}{2+3} = 0.4$

$\mathrm{x_2} = \frac{{}^{\mathrm{n}}\mathrm{O_2}}{\mathrm{n_2}+{}^{\mathrm{n}}\mathrm{O_2}} = \frac{3}{2+3} = 0.6$

$\therefore {}^{\mathrm{P}}\mathrm{N_2} = 0.4 \times 10 = 4 \text{ atm}, {}^{\mathrm{P}}\mathrm{O_2} = 0.6 \times 10 = 6 \text{ atm}$

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Q.15 Answer:

4

Solution:

$\frac{1}{\sqrt{MH_2}}: \frac{1}{\sqrt{MO_2}}: \frac{1}{\sqrt{MH_2}}$

After 4 steps

$\frac{1}{\sqrt{2}}^4 : \frac{1}{\sqrt{3^2}}^4 : \frac{1}{\sqrt{4}}^4$

$\frac{1}{2\times 2} : \frac{1}{32\times 32} : \frac{1}{4\times 4}$

$16\times 16 : 1 : 8\times 8\times 1 \quad : y$

$\frac{Y}{T} = 4$

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